A separable route to the dRPA density–density moments under tensor hypercontraction

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Abstract

The direct random-phase approximation supplies the density–density moments that moment-conserving *GW* convolves into a self-energy. Under density fitting those moments are reached by contracting a three-index vertex against the particle–hole energy denominator; under tensor hypercontraction (THC) the vertex factorises further, but the denominator does not, and every previous attempt in this project handled the mismatch by partitioning the particle–hole pairs and treating each region with a different expansion. This note derives a route that removes the partition. The denominator is represented on a grid of exponentials shared by every pole of the certified rational approximation to $\eta^{(0)}$ and by every moment order, the powers of the denominator are recovered exactly by a binomial expansion in the orbital energies, and a single measured scalar residual bounds the relative spectral error of the resulting auxiliary-space objects at *every* order. Three candidate treatments of the higher powers are compared against an exact evaluation; two are rejected on measurement. The cost model and the measured THC rank requirement are given: every stage is cubic in operations and quadratic in memory, which is the property the whole exercise exists to obtain, while the constants place the crossover against the density-fitted route near one thousand orbitals.

The requirement

Every stage of a tensor-hypercontracted path must be $O(N^3)$ in operations and $O(N^2)$ in memory. Nothing else justifies the construction: the density-fitted route is already available, is simpler, and carries a certified error budget, so a hypercontracted route that is merely faster by a constant has no reason to exist. What it buys is the exponent.

The distinction between constant and exponent is worth stating sharply, because the two have different owners. The constant is dominated by the hypercontraction rank, a property of the factorisation algorithms, which are an active field and are expected to improve. The exponent is a property of the solver. No improvement in the integrals can repair a quartic contraction. A poor constant is therefore a reason to wait; a poor exponent is a defect, and the scaling of each stage is reported below on that basis rather than as an afterthought.

1 The object to be computed

Write the particle–hole gaps as $D_{ia} = \epsilon_a - \epsilon_i > 0$ and let $\mathbf{V} \in \mathbb{R}^{N_{\mathrm{aux}} \times N_{ov}}$ be the vertex, so that the occupied–virtual electron repulsion integrals are

$$(ia|jb) = \sum_P V_{P,ia} V_{P,jb}. \tag{1}$$

The moments of the density–density response, projected into the auxiliary space, are

$$\bar{\boldsymbol{\eta}}^{(m)} = \mathbf{V}\boldsymbol{\eta}^{(m)}\mathbf{V}^T, \qquad \boldsymbol{\eta}^{(m)} = (\mathbf{X}+\mathbf{Y})\,\boldsymbol{\Omega}^m(\mathbf{X}+\mathbf{Y})^T, \tag{2}$$

with $\boldsymbol{\Omega}$ the RPA excitation energies and $\mathbf{X}$, $\mathbf{Y}$ the RPA eigenvectors. These are the quantities the self-energy convolution consumes. The symbols $\mathbf{X}$ and $\mathbf{Y}$ appear only in this equation; from Section 2 onwards X denotes the THC collocation matrix.

Two families of auxiliary-space objects generate all of them. The first is purely polynomial in the gaps,

$$\mathcal{A}^{(m)} = \mathbf{V}\,\mathbf{D}^m\,\mathbf{V}^T \qquad \text{i.e.} \qquad \mathcal{A}^{(m)}_{PQ} = \sum_{ia} V_{P,ia}\,D^m_{ia}\,V_{Q,ia}, \tag{3}$$

and the second carries the RPA screening,

$$\mathcal{N}^{(m)} = \frac{2}{\pi}\int_0^\infty \Big(\mathbf{I} + 4\mathcal{Q}^{(1)}(t)\Big)^{-1}\,\mathcal{Q}^{(m+1)}(t)\,\mathrm{d}t, \qquad \mathcal{Q}^{(k)}(t) = \mathbf{V}\,\frac{\mathbf{D}^k}{\mathbf{D}^2+t^2}\,\mathbf{V}^T. \tag{4}$$

Equation (4) is the compact form of the RPA integral: it follows from writing $\boldsymbol{\eta}^{(0)} = (\mathbf{A}+\mathbf{B})^{-1}[(\mathbf{A}+\mathbf{B})(\mathbf{A}-\mathbf{B})]^{1/2}$, using $\mathbf{M}^{1/2} = {}^2\!/_\pi\,\mathbf{M}\int_0^\infty (t^2\mathbf{I}+\mathbf{M})^{-1}\mathrm{d}t$, applying the Woodbury identity, and then eliminating the bare term with the push-through identity $(\mathbf{I}+\mathbf{M})^{-1} = \mathbf{I} - \mathbf{M}(\mathbf{I}+\mathbf{M})^{-1}$. Both terms that would otherwise have to cancel against one another disappear analytically.

The two families combine into every moment through one recursion. Writing

$$\mathcal{A}^{(m)} \text{ for odd } m, \qquad \mathcal{N}^{(m)} \text{ for even } m, \tag{5}$$

and calling either $\mathbf{A}^{(m)}$,

$$\bar{\boldsymbol{\eta}}^{(m)} = \mathbf{A}^{(m)} + 4\sum_{\substack{k=1,3,5,\ldots\\ k\le m-1}} \mathbf{A}^{(m-1-k)}\,\bar{\boldsymbol{\eta}}^{(k)}, \qquad \bar{\boldsymbol{\eta}}^{(0)} = \mathcal{N}^{(0)}, \quad \bar{\boldsymbol{\eta}}^{(1)} = \mathcal{A}^{(1)}. \tag{6}$$

The parity of $m-1-k$ always matches that of m, so the odd moments are built entirely from $\mathcal{A}$ and the even ones entirely from $\mathcal{N}$. This is an $O(N^3_{\text{aux}})$ recursion that never touches the particle–hole index.

1.1 The rational form of the integral

In production the t-integral in (4) is not quadratured. The inverse square root is replaced by a certified rational approximation $x^{-1/2} \approx \sum_j w_j(x+s_j)^{-1}$ on a rigorous spectral enclosure of $\tilde{\mathbf{M}} = \mathbf{D}^2 + 4\mathbf{W}\mathbf{W}^T$, $\mathbf{W} = \mathbf{D}^{1/2}\mathbf{V}^T$, whose poles s_j and weights w_j are chosen against a *measured* scalar error. Equation (4) becomes a finite sum,

$$\mathcal{N}^{(m)} = \sum_j w_j\,\Big(\mathbf{I} + 4\mathcal{Q}^{(1)}(s_j)\Big)^{-1}\,\mathcal{Q}^{(m+1)}(s_j), \qquad \mathcal{Q}^{(k)}(s) = \mathbf{V}\,\frac{\mathbf{D}^k}{\mathbf{D}^2+s}\,\mathbf{V}^T, \tag{7}$$

with the ${}^2\!/_\pi$ absorbed into w_j. At the production tolerance of 10^{-14} this is 18 to 19 poles on the systems used below. Everything that follows targets (7): the shifts are a fixed, known set rather than a grid whose scale has to be optimised per calculation.

2 Why tensor hypercontraction does not simply substitute

Under THC the vertex factorises across the orbital pair,

$$(ia|jb) = \sum_{PQ} X_{iP} X_{aP}\, Z_{PQ}\, X_{jQ} X_{bQ}, \tag{8}$$

so the natural replacement for $\mathbf{V}$ is $\mathbf{L}^T\mathbf{X}$, where $X_{P,ia} = X_{iP}X_{aP}$ and $\mathbf{Z} = \mathbf{L}\mathbf{L}^T$. Substituting into (3) costs nothing. Referencing the orbital energies to the chemical potential μ — writing $\tilde{\epsilon}_i = \epsilon_i - \mu$ and $\tilde{\epsilon}_a = \epsilon_a - \mu$, which leaves $D_{ia} = \tilde{\epsilon}_a - \tilde{\epsilon}_i$ unchanged — the power expands binomially into terms separable in i and a,

$$\mathcal{A}^{(m)} \to \mathbf{L}^T \mathcal{A}^{(m)}_X \mathbf{L}, \qquad \mathcal{A}^{(m)}_{X,PQ} = \sum_{p=0}^{m} \binom{m}{p} \left[\sum_i X_{iP} X_{iQ} (-\tilde{\epsilon}_i)^p\right] \left[\sum_a X_{aP} X_{aQ} \tilde{\epsilon}_a^{m-p}\right], \tag{9}$$

which is $O(N^3)$ and is how the THC dTDA already works. The shift is not needed here — no exponentials are involved — but it costs nothing and keeps the expansion identical to the one Section 4 needs.

Substituting into (7) does not work, because $\mathbf{D}^k/(\mathbf{D}^2 + s)$ is not separable. Every earlier attempt in this project addressed that by splitting the particle–hole pairs at each quadrature point z into $D \ll z$, $D \sim z$ and $D \gg z$ regions, using the asymptotic series $D^{2r+1}/(D^2+z^2) = \sum_c (-1)^c D^{2r+1+2c} z^{-2c-2}$ on one side, the Laplace/cosine integral $\int_0^\infty \cos(zp)\, e^{-Dp}\, \mathrm{d}p$ on the other, and correction terms across the boundary. The separable builders can only mask on i and a independently, so the region — a staircase in the sorted (i, a) grid — has to be approximated by unions of rectangles, and the bookkeeping dominates.

That partition is *not* required by the factorisation. It is required by the oscillatory inner quadrature, which cannot resolve $\cos(zp)$ against e^{-Dp} across a wide range of D at a single scale. Replacing the quadrature removes the partition.

3 A shared exponential representation

3.1 The representation

With the same referencing to μ, so that $\tilde{\epsilon}_i < 0 < \tilde{\epsilon}_a$, on a set of positive exponents $\{t_k\}$,

$$g_s(D) \equiv \frac{D}{D^2 + s} \;\approx\; \sum_k c_k(s)\, e^{-t_k D}, \qquad e^{-t_k D} = e^{t_k \tilde{\epsilon}_i}\, e^{-t_k \tilde{\epsilon}_a}. \tag{10}$$

The shift enters only through the coefficients: **the exponents are shared by every pole and by every moment order**, so whatever a solver builds per exponent is built once. The μ reference is load-bearing rather than cosmetic — it puts both factors in $[0, 1]$, so neither overflows however wide the orbital range, and it makes the binomial in (9) a sum of like-signed terms.

3.2 What the residual certifies

The coefficients are obtained by weighted least squares against the *true* set of gaps, minimising $\sum_{ia}\left[(\sum_k c_k e^{-t_k D_{ia}} - g_s(D_{ia}))/g_s(D_{ia})\right]^2$, so the reported residual

$$\varepsilon = \max_{ia} \frac{|\delta(D_{ia})|}{g_s(D_{ia})}, \qquad \delta(D) = \sum_k c_k e^{-t_k D} - g_s(D), \tag{11}$$

is realised on the system at hand rather than bounded over an enclosing interval. It then transfers directly to the operator, and — this is the point — it does so at every moment order. Since $\Delta\mathcal{Q}^{(m+1)} = \mathbf{V}\,\mathrm{diag}(\mathbf{D}^m\delta)\,\mathbf{V}^T$ is symmetric, for any unit $\mathbf{u}$

$$\left|\mathbf{u}^T\Delta\mathcal{Q}^{(m+1)}\mathbf{u}\right| \;=\; \left|\sum_{ia} D_{ia}^m\,\delta(D_{ia})\,(\mathbf{V}^T\mathbf{u})_{ia}^2\right| \;\le\; \varepsilon\sum_{ia} D_{ia}^m\,g_s(D_{ia})\,(\mathbf{V}^T\mathbf{u})_{ia}^2 \;=\; \varepsilon\,\mathbf{u}^T\mathcal{Q}^{(m+1)}\mathbf{u}, \tag{12}$$

because $D^m g_s(D) > 0$ makes $\mathcal{Q}^{(m+1)}$ positive semidefinite. Hence

$$\left\|\Delta\mathcal{Q}^{(m+1)}\right\|_2 \;\le\; \varepsilon\left\|\mathcal{Q}^{(m+1)}\right\|_2 \qquad \text{for every } m. \tag{13}$$

One scalar number certifies the whole moment set, and it does not degrade with order.

3.3 Selecting the exponents

The exponential basis on a real gap distribution has a numerical rank near 30 in double precision: the least-squares effective rank saturates between 27 and 34 whether a geometric grid holds 48 points or 120. A geometric layout therefore reaches the achievable floor only by oversampling. Ranking a dense set of candidates by pivoted QR on the column-normalised basis $[e^{-tD_{ia}}]$ and keeping the leading pivots reaches the same floor with far fewer points (Table 1).

	worst relative residual on g_s, all poles	
points	geometric	pivot-selected
36	4.4×10^{-8}	1.4×10^{-10}
42	4.5×10^{-9}	$\mathbf{1.0\times10^{-12}}$
60	1.9×10^{-11}	2.9×10^{-12}

Table 1: Benzene, cc-pVDZ, 18 poles at a requested $\eta^{(0)}$ tolerance of 10^{-14}. Solver cost is linear in the number of exponents, so the saving is direct.

Below a few times 10^{-13} the fit is limited by double precision rather than by the grid, and more points do not help; a request below that is rejected rather than silently answered. Table 2 gives the sizes the search returns.

4 The higher powers

Equation (10) supplies $\mathcal{Q}^{(1)}$. The moments need $\mathcal{Q}^{(m+1)}$ for even m up to $n_{\max}-1$, and there are three ways to reach them.

	exponents required		
system	$\varepsilon \leq 10^{-6}$	10^{-9}	10^{-11}
lithium hydride	16	19	21
water	22	28	32
ozone	23	28	36
benzene	25	33	40

Table 2: cc-pVDZ, 18 to 19 poles. The measured residual is the realised one on the system's own gaps, and the operator error it certifies through (13) runs one to two decades inside it.

A: polynomial division. The identity

$$\frac{D^{2r+1}}{D^2+s} = \sum_{k=0}^{r-1} (-s)^k D^{2r-1-2k} \;+\; (-s)^r \frac{D}{D^2+s} \tag{14}$$

is exact for every pair, and every term but the last is a shift-independent $\mathcal{A}^{(j)}$ that the dTDA already builds. Contracted into the auxiliary space this needs only two builds per exponent and one set of $\mathcal{A}$ for the whole calculation, so it is much the cheapest. It is also, in disguise, what the previous implementations' `build_large` plus `build_correction_large` amounted to.

B: binomial in the orbital energies. Keep the fit at $m = 0$ and recover the power exactly, as in (9):

$$\mathcal{Q}_X^{(m+1)}(s) \;\approx\; \sum_k c_k(s) \sum_{p=0}^{m} \binom{m}{p} \mathbf{O}_p(t_k) \circ \mathbf{U}_{m-p}(t_k), \tag{15}$$

with $\circ$ the elementwise product and

$$\big[\mathbf{O}_p(t)\big]_{PQ} = \sum_i X_{iP} X_{iQ} \, (-\tilde{\epsilon}_i)^p \, e^{t\tilde{\epsilon}_i}, \qquad \big[\mathbf{U}_q(t)\big]_{PQ} = \sum_a X_{aP} X_{aQ} \, \tilde{\epsilon}_a^q \, e^{-t\tilde{\epsilon}_a}. \tag{16}$$

Every factor is non-negative in sign structure, so no cancellation is introduced. The cost is $(n_{\max}+1)$ builds per exponent instead of two.

C: fit the target directly. Fit $D^{m+1}/(D^2+s)$ on the same exponents and keep two builds per exponent at every order.

Route A loses about one and a half decades per moment order. The terms of (14) carry factors of s^k that the answer does not, and although the cancellation is benign where $s \lesssim D_{\max}^2$, it is not where the pole shift is much larger. A finer grid does not recover it, because the loss is not in the fit.

Route C saturates: at 100 exponents it is no better than at 40. A function growing like D^{m-1} cannot be built from decaying exponentials without cancellation that double precision will not carry.

Route B is flat in the moment order at 10^{-12}, exactly as (13) predicts, and is the route adopted. It is worth noting that this kernel — the binomial expansion of $D^m e^{-tD}$ — is the one the earlier implementations already contained; what changes is everything around it.

	relative error on $\mathcal{N}^{(m)}$, max norm		
m	A: division	B: binomial	C: direct fit
0	5.3×10^{-13}	5.3×10^{-13}	5.3×10^{-13}
2	9.7×10^{-11}	9.7×10^{-13}	6.8×10^{-8}
4	1.0×10^{-8}	1.0×10^{-12}	2.3×10^{-4}
6	1.1×10^{-6}	1.0×10^{-12}	1.2×10^{-1}
8	$\mathbf{1.2 \times 10^{-4}}$	$\mathbf{9.8 \times 10^{-13}}$	$\mathbf{9.3 \times 10^{-1}}$

Table 3: Benzene, cc-pVDZ, 42 exponents, measured against an exact evaluation of the same pole sum (7) with a density-fitted vertex, so that only the treatment of the denominator is under test.

5 Removing the Cholesky factor

Substituting $\mathbf{V} \to \mathbf{L}^T\mathbf{X}$ into (7) would appear to require $\mathbf{L}$, a Cholesky factor of $\mathbf{Z}$, since $\mathcal{Q}^{(k)} = \mathbf{L}^T\mathcal{Q}_X^{(k)}\mathbf{L}$ with $\mathcal{Q}_X^{(k)} = \mathbf{X}[\mathbf{D}^k/(\mathbf{D}^2+s)]\mathbf{X}^T$. The push-through identity $(\mathbf{I}+\mathbf{AB})^{-1}\mathbf{A} = \mathbf{A}(\mathbf{I}+\mathbf{BA})^{-1}$ removes it:

$$\left(\mathbf{I}+4\mathbf{L}^T\mathcal{Q}_X^{(1)}\mathbf{L}\right)^{-1}\mathbf{L}^T = \mathbf{L}^T\left(\mathbf{I}+4\mathcal{Q}_X^{(1)}\mathbf{L}\mathbf{L}^T\right)^{-1} = \mathbf{L}^T\left(\mathbf{I}+4\mathcal{Q}_X^{(1)}\mathbf{Z}\right)^{-1}, \tag{17}$$

so that $\mathcal{N}^{(m)} = \mathbf{L}^T\mathcal{N}_X^{(m)}\mathbf{L}$ with

$$\mathcal{N}_X^{(m)} = \sum_j w_j\left(\mathbf{I}+4\mathcal{Q}_X^{(1)}(s_j)\mathbf{Z}\right)^{-1}\mathcal{Q}_X^{(m+1)}(s_j). \tag{18}$$

The price is a non-symmetric solve in place of a Cholesky one, at the same $O(N_{\mathrm{THC}}^3)$. The same substitution turns (6) into

$$\boldsymbol{\zeta}^{(m)} = \mathbf{A}_X^{(m)} + 4\sum_{\substack{k=1,3,5,\ldots\\ k\le m-1}}\mathbf{A}_X^{(m-1-k)}\,\mathbf{Z}\,\boldsymbol{\zeta}^{(k)}, \qquad \bar{\boldsymbol{\eta}}^{(m)} = \mathbf{L}^T\boldsymbol{\zeta}^{(m)}\mathbf{L}, \tag{19}$$

with $\mathbf{A}_X^{(m)}$ equal to $\mathcal{A}_X^{(m)}$ for odd m and $\mathcal{N}_X^{(m)}$ for even m. The $\boldsymbol{\zeta}$ produced here is in the same convention the existing THC dTDA self-energy builder consumes, which applies the two outer factors of $\mathbf{Z}$ at contraction time; no interface change is needed downstream.

6 Algorithm and cost

Counting the passes each side actually takes, the leading terms are

$$\text{separable:} \quad n_t(n_{\max}+1)N_{\mathrm{THC}}^2N \;+\; \tfrac{1}{2}(n_{\max}+1)^2N_{\mathrm{THC}}^3, \tag{20}$$

$$\text{density-fitted:} \quad (3n_{\mathrm{pole}}+2n_{\max})\,N_{\mathrm{aux}}^2\,N_{ov}. \tag{21}$$

The separable route is cubic in the system size and the density-fitted route quartic, so there is a crossover; where it falls is set almost entirely by the THC rank, which enters squared. Table 4 evaluates the model at benzene's shape ratios.

Algorithm 1 THC dRPA density–density moments

Reference the orbital energies to μ; form the gaps D_{ia}.
Bound the spectrum of $\tilde{\mathbf{M}}$; select poles $\{s_j, w_j\}$ against the requested $\eta^{(0)}$ tolerance.
Rank candidate exponents by pivoted QR; grow the grid until the measured residual ε meets its tolerance; fit $c_k(s_j)$ for every pole. $O(N_{ov} n_t^2)$
for each exponent t_k **do**
    build $\mathbf{O}_p(t_k)$, $\mathbf{U}_q(t_k)$ for $p, q \le n_{\max}$ $O(n_{\max} N_{\mathrm{THC}}^2 N)$
    $\mathbf{R}_m(t_k) \leftarrow \sum_p \binom{m}{p} \mathbf{O}_p \circ \mathbf{U}_{m-p}$ for even m
end for
$\mathcal{Q}_X^{(m+1)}(s_j) \leftarrow \sum_k c_k(s_j) \mathbf{R}_m(t_k)$ $O(n_{\mathrm{pole}} n_{\max} n_t N_{\mathrm{THC}}^2)$
$\mathcal{N}_X^{(m)} \leftarrow \sum_j w_j (\mathbf{I} + 4\mathcal{Q}_X^{(1)}(s_j)\mathbf{Z})^{-1} \mathcal{Q}_X^{(m+1)}(s_j)$
$\mathcal{A}_X^{(m)}$ for odd m by (9), without the exponential factors.
$\boldsymbol{\zeta}^{(m)}$ by the recursion (19). $O(n_{\max}^2 N_{\mathrm{THC}}^3)$

The self-energy stage that consumes $\boldsymbol{\zeta}^{(m)}$ has to meet the same requirement, and in a direct implementation it does not. Written as a loop over the Green's function orbitals it contracts an (N_{THC}, N) vertex against $\boldsymbol{\zeta}$ once per orbital, which is $O(N_{\mathrm{THC}}^2 N^2)$ operations over an $O(N^3)$ array. The orbital index reaches the output only through the weighted sums the moment convolution takes, so moving that sum inside the auxiliary contraction gives

$$\Sigma^{(n)} = \sum_{t \le n} \binom{n}{t} (\pm 1)^t \, \mathbf{X}^T \left(\boldsymbol{\zeta}'^{(t)} \circ \mathbf{G}^{(n-t)} \right) \mathbf{X}, \qquad G_{PQ}^{(j)} = \sum_{x \in \mathrm{sector}} \epsilon_x^j X_{xP} X_{xQ}, \tag{22}$$

with the sign negative for the occupied sector and the whole t sum accumulated in the auxiliary space before a single rotation per order. Neither the array nor the loop survives, and the stage becomes $O(n_{\max}^2 (N_{\mathrm{THC}}^2 N + N_{\mathrm{THC}} N^2))$ operations in $O(n_{\max}(N_{\mathrm{THC}}^2 + N^2))$ memory.

	crossover, orbitals		
THC rank	$n_t = 24$	$n_t = 33$	$n_t = 42$
$5 \times N$	114	134	155
$8 \times N$	380	433	486
$10 \times N$	686	769	853
$12 \times N$	1121	1241	1361

Table 4: $n_{\max} = 9$, 18 poles at an $\eta^{(0)}$ tolerance of 10^{-14}, benzene's auxiliary-to-orbital and pair-to-orbital ratios held fixed.

7 Error budget

Three approximations enter, and each is measured separately rather than inferred from the others.

1. **The pole set.** Certified by the measured scalar error of the rational approximation to $x^{-1/2}$ over a rigorous enclosure of $\tilde{\mathbf{M}}$; at the production default this is $\sim 10^{-15}$ on 18 to 19 poles.

2. **The separable representation.** Certified by ε through (13), at every moment order. Measured, the realised operator error runs one to two decades inside the bound.

3. **The factorisation itself.** Not certified — measured. This is the dominant term in practice and it is discussed below.

The factorisation error should be quoted against the *density-fitted* integrals, not the exact ones: least-squares THC fits the density-fitted tensor, and on water in cc-pVDZ the auxiliary basis contributes 4.4×10^{-3} relative on its own, which is larger than anything the factorisation adds and hides it completely.

Table 5 gives the measured relationship for the most efficient construction available. The frontier shift is quoted for the worst of the systems tested at each accuracy rather than the average, because the systems do not fail together: at a rank of $8 \times N$ water and ozone are already inside 6 meV while lithium hydride is out at 88 meV, its diffuse virtual orbital being the first thing the fit loses.

rank / N	relative Frobenius error	worst frontier shift
4.6 – 5.7	2.8×10^{-2} – 6.9×10^{-2}	278 meV
7.3 – 8.9	2.1×10^{-3} – 3.6×10^{-3}	88 meV
8.1 – 10.7	6.4×10^{-4} – 1.0×10^{-3}	7.7 meV
8.7 – 11.4	1.8×10^{-4} – 3.5×10^{-4}	2.6 meV
9.0 – 13.6	4.5×10^{-5} – 1.8×10^{-4}	1.1 meV

Table 5: Least-squares THC with randomly-pivoted Cholesky point selection, cc-pVDZ, across water, lithium hydride and ozone; error against the density-fitted tensor in the same auxiliary basis, frontier shift against the density-fitted dTDA at $n_{\max} = 3$.

A few meV therefore costs a rank near $11 \times N$, and the requirement does not fall with system size: benzene at $N = 114$ reaches -2.5 meV at $9.96 \times N$, the same band as the small systems. Read together with Table 4, that places the crossover against the density-fitted route near a thousand orbitals. The exponent grid is already close to its own floor, so neither of the two levers has much left to give.

8 Summary

The particle–hole denominator separates cleanly once it is represented on a shared exponential grid fitted per pole against the true gaps, with the powers of the denominator recovered exactly by a binomial expansion in the orbital energies rather than fitted. The region partitioning that dominated earlier implementations is an artefact of the oscillatory inner quadrature, not of the factorisation, and it disappears with it. A single measured scalar residual certifies the relative spectral error of every auxiliary-space object at every moment order, and the Cholesky factor of the THC Coulomb matrix is removable by a push-through identity.

Every stage is then cubic in operations and quadratic in memory, which is the property the construction exists to obtain.

The constants are another matter. Solver cost scales with the square of the hypercontraction rank; the rank needed for few-meV quasiparticle agreement is near $11 \times N$ and does not fall with system

size; and the resulting crossover against the density-fitted route sits near one thousand orbitals. That is a statement about today's factorisation algorithms rather than about this construction, and it is the half of the problem that improvement in the integrals can reach. The exponent is the half that it cannot, and it is now correct.